



Building Trust in Autonomous Vehicles

Addressing Public Concern About AI-Driven Cars

Voltaris Mobility N.V. • Amsterdam, Netherlands • Founded 2016

“Drive the Current.”

The Trust Problem

Perhaps the single biggest obstacle to autonomous vehicles is not technology but trust. Many people are uneasy about the idea of a car that drives itself, and that hesitation is rational — handing control to a machine is a profound shift. Voltaris treats public trust as a core engineering and communication challenge, not an afterthought.

Why People Are Concerned

- **Loss of control** — discomfort with letting software make driving decisions.
- **Safety doubts** — fear that the system will fail in an unusual or dangerous situation.
- **Lack of transparency** — not understanding how or why the car makes its choices.
- **High-profile incidents** — rare accidents involving autonomous systems receive intense coverage.
- **Accountability** — uncertainty about who is responsible when something goes wrong.

How Voltaris Builds Confidence

The company's approach is to earn trust gradually and visibly, rather than asking for it all at once. Several principles guide this effort:

- **Gradual rollout** — autonomy is introduced in tiers (Halo Assist, Pilot, Autonomy), letting drivers build familiarity step by step.
- **Geofenced Level 4** — full self-driving operates only in carefully mapped and validated zones, never claiming more capability than has been proven.
- **Transparency** — the in-car display shows what the system sees and intends to do, so passengers understand its decisions.
- **Safety record reporting** — Voltaris publishes safety data and intervention rates rather than hiding them.
- **Always a way out** — supervised modes keep the human in control, and the robotaxi service offers in-app human support at any time.

Designing for Reassurance

Trust is also a design problem. In the Helix robotaxi, the cabin is deliberately calm and transparent: passengers can see a clear visualisation of the vehicle's perception of the road, smooth and predictable driving behaviour is prioritised over aggressive efficiency, and a visible support button connects riders to a human operator instantly. The aim is for the first ride to feel unremarkable — the strongest possible endorsement.

Education & Demonstration

Voltaris invests in letting people experience autonomy safely before they rely on it — through supervised test drives, public robotaxi pilots in partner cities, and clear educational materials

explaining what the system can and cannot do. The company's view is that nothing builds trust like a safe, uneventful ride, and that scaling carefully is itself the most effective trust strategy.

Accountability & Regulation

Finally, Voltaris engages openly with regulators and is explicit about responsibility: in supervised modes the driver remains in control, while in fully autonomous Level 4 operation the company stands behind the system. Clear accountability, the company argues, is essential to public confidence in the technology.